

EC442: Lecture Notes

Chen Gao

Notes on the 1st year Macro class

September 29, 2026

Contents

1. Lecture 1 Sep. 29	3
1.1. Why heterogeneity	3
1.2. How to model incomplete markets?	3
1.3. Planning problem	4
1.4. Towards a recursive formulation	5
1.5. Dynamic Programming	6
1.6. Contraction mapping theorem	6
1.6.1. Application to planner	7
1.7. Using the value function analytically	7
1.8. Using the value function numerically	7
Bibliography	10

1. Lecture 1 Sep. 29

1.1. Why heterogeneity

Complete markets + standard welfare assumptions:

- Competitive eq. Pareto efficient; planner representation w/ fixed weights $\alpha_i > 0$.
- Common discounting; preferences $u(c) - v(n)$. Consumption FOCs:

$$\alpha_i u'(c_t^i) = \alpha_j u'(c_t^j) \Rightarrow \frac{u'(c_t^i)}{u'(c_t^j)} = \frac{\alpha_j}{\alpha_i}. \quad (1.1)$$

Marginal-utility ratio constant across time/states.

- Common CRRA: $u'(c) = c^{-\sigma}$.

$$\frac{c_t^i}{c_t^j} = \left(\frac{\alpha_i}{\alpha_j} \right)^{\frac{1}{\sigma}}. \quad (1.2)$$

Fixed consumption shares $\rightarrow$ **perfect insurance** against idiosyncratic income shocks. Aggregate risk remains; unequal Pareto weights $\rightarrow$ unequal consumption.

- Strictly convex labor disutility; interior labor choice; aggregate conditions fixed: productivity $\uparrow \rightarrow$ labor $\uparrow$; consumption share unchanged.

Implications:

- Karl Marx
- Full insurance at odds w/ data

Lecture addition: Intuition for Complete Markets

- **Example:** 18 students; one high earner, others lower. Complete markets $\rightarrow$ high earner sells state-contingent claims $\rightarrow$ consumption decoupled from income.
- **Marx analogy:** "From each according to ability, to each according to needs." Efficient allocation equalizes marginal utilities regardless of income.
- **Empirical mismatch:** Data shows strong income-consumption correlation. Complete markets imply correlation vanishes $\rightarrow$ model unrealistic. Heterogeneity requires market incompleteness.

Recording: 00:13:54, 00:16:04

1.2. How to model incomplete markets?

1. **Exogenous:** assume some markets absent.
 1. e.g. save/borrow via noncontingent bonds; no state-contingent claims.
2. **Endogenous:** derive market incompleteness from frictions.
 1. e.g. limited commitment (participation constraints), private information.
 2. deeper but harder to do

Lecture addition: Course Roadmap

- **Methods:** Dynamic programming, numerical methods.
- **Background:** Equilibrium and welfare theorems.
- **Growth:** Basic models of economic growth.
- **Incomplete markets:** Endogenous (participation constraints/limited commitment), exogenous (asset structure), search models.

Recording: 00:31:25, 00:33:02

Lecture addition: Frictions for Endogenous Incompleteness

- **Limited commitment:** Contracts unenforceable. E.g., US car insurance limits; agents flee jurisdiction (e.g., to Frankfurt) to avoid liability → limits risk trading.
- **Private information:** Effort/talent unobservable. Income sharing → moral hazard (effort reduction) → complete insurance inefficient/impossible.

Recording: 00:22:10, 00:27:38

1.3. Planning problem

Set $n_t = A = 1$; $f(k) = F(k, 1)$. Given $k_0 \geq 0$. Discount factor $0 < \beta < 1$; depreciation $0 \leq \delta \leq 1$.

Choose sequences $c_t, k_{t+1} \geq 0$:

$$\max \sum_{t=0}^{\infty} \beta^t u(c_t) \text{ s.t. } c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t. \quad (1.3)$$

- **Lagrangian;** current-value multipliers λ_t :

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t [u(c_t) + \lambda_t (f(k_t) + (1 - \delta)k_t - c_t - k_{t+1})]. \quad (1.4)$$

- **Interior FOCs:**

$$\begin{aligned} c_t : u'(c_t) &= \lambda_t, \\ k_{t+1} : \lambda_t &= \beta \lambda_{t+1} [f'(k_{t+1}) + 1 - \delta]. \end{aligned} \quad (1.5)$$

Euler equation:

$$u'(c_t) = \beta u'(c_{t+1}) [f'(k_{t+1}) + 1 - \delta]. \quad (1.6)$$

- **Transversality condition:**

$$\lim_{T \rightarrow \infty} \beta^T \lambda_T k_{T+1} = \lim_{T \rightarrow \infty} \beta^T u'(c_T) k_{T+1} = 0. \quad (1.7)$$

Discounted shadow value of terminal capital $\rightarrow 0$.

- Increasing, concave u ; concave f ; finite discounted utility: feasibility + FOCs + TVC $\rightarrow$ global optimum. At corners, use KKT conditions.

1.4. Towards a recursive formulation

- Sequence approach: infinitely many dated choices + optimality conditions.
- Repeated trade-off: consume today vs. save for tomorrow.
- Stationary problem; current capital k_t summarizes relevant history.
- Recursion: continuation value summarizes future choices; same decision problem at every date.

Value function: maximal lifetime utility from initial capital k_0 :

$$v^*(k_0) = \max_{\{c_t, k_{t+1}\}_{t \geq 0}} \sum_{t=0}^{\infty} \beta^t u(c_t). \quad (1.8)$$

Feasible sequences: $c_t, k_{t+1} \geq 0$; $c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t$, all $t \geq 0$.

Split date 0 from continuation:

$$v^*(k_0) = \max_{c_0, k_1} \left[u(c_0) + \beta \max_{\{c_t, k_{t+1}\}_{t \geq 1}} \sum_{t=1}^{\infty} \beta^{t-1} u(c_t) \right]. \quad (1.9)$$

- Outer max: $c_0, k_1 \geq 0$; $c_0 + k_1 = f(k_0) + (1 - \delta)k_0$.
- Inner max: given chosen k_1 , same feasibility constraints for all $t \geq 1$. Discounting restarts at date 1 $\rightarrow$ inner value $v^*(k_1)$.

Continuation value absorbs future choices. **Bellman equation** (current capital k , next-period capital k'):

$$v^*(k) = \max_{c, k'} [u(c) + \beta v^*(k')] \quad (1.10)$$

subject to

$$c + k' = f(k) + (1 - \delta)k, c, k' \geq 0. \quad (1.11)$$

Infinite sequence $\rightarrow$ one-period choice + discounted continuation value. Stationarity $\rightarrow$ same problem at every date.

Bellman equation = **functional equation**; unknown = function $v(k)$.

Questions:

- **Existence:** conditions for solution v ?
- **Uniqueness:** conditions for unique v within chosen function class?
- **Equivalence:** when does $v = v^*$, with maximizing policy solving original sequence problem?

Answer:

- *Recursive Macroeconomic Theory* (RMT), “Dynamic Programming” ([Ljungqvist & Sargent, 2018, ch. 3](#)).

Lecture addition: Computational Advantage of DP

- **Lagrangian**: Leads to 2nd-order difference eq. Solving via “shooting” → local, clunky.
- **Recursive**: Bellman eq. → direct computation of **global solution** via iteration.
- **Convergence**: VFI guaranteed to converge to unique fixed point (via CMT), unlike many ODE/PDE methods.

Recording: 01:06:37, 01:07:39

1.5. Dynamic Programming

Bellman operator T : candidate continuation value $v \rightarrow$ updated value Tv .

$$(Tv)(k) = \max_{c,k'} [u(c) + \beta v(k')] \quad (1.12)$$

subject to

$$c + k' = f(k) + (1 - \delta)k, c, k' \geq 0. \quad (1.13)$$

Bellman solution = **fixed point**:

$$Tv = v \Leftrightarrow (Tv)(k) = v(k) \text{ for all feasible } k. \quad (1.14)$$

Fixed-point theorems give existence/uniqueness conditions within chosen function space; assumptions matter.

1.6. Contraction mapping theorem

Theorem 1.6.1 (Contraction mapping theorem): Let (S, d) be a nonempty complete metric space and $T : S \rightarrow S$ a contraction: there exists $\beta \in (0, 1)$ such that

$$d(Tx, Ty) \leq \beta d(x, y) \text{ for all } x, y \in S. \quad (1.15)$$

Then T has a unique fixed point $v \in S$. For every $v_0 \in S$ and $n = 0, 1, 2, \dots$,

$$d(T^n v_0, v) \leq \beta^n d(v_0, v). \quad (1.16)$$

Hence $T^n v_0 \rightarrow v$ as $n \rightarrow \infty$, with geometric convergence. Here T^n denotes n successive applications of T , with T^0 the identity.

Lecture addition: Intuition for Contraction Mapping

- Analogous to Brouwer fixed-point theorem in 1D: continuous function mapping interval to itself with slope $< 1 \rightarrow$ unique fixed point.
- In function spaces, “slope” = contraction factor β . Iterating operator moves points closer $\rightarrow$ convergence to unique fixed point.

Recording: 00:59:06, 01:01:46

1.6.1. Application to planner

- Assume feasible capital remains in nonempty compact interval K .
- $S = C(K)$: continuous real-valued functions on K ; automatically bounded.
- **Sup-norm metric**:

$$d(v, w) = \|v - w\|_\infty = \sup_{k \in K} |v(k) - w(k)|. \quad (1.17)$$

(S, d) complete.

- Must verify $T : S \rightarrow S$: updated values remain bounded and continuous. Then Bellman discounting gives

$$\|Tv - Tw\|_\infty \leq \beta \|v - w\|_\infty. \quad (1.18)$$

Theorem 1.6.1 $\rightarrow$ unique fixed point in S ; value iteration converges from any $v_0 \in S$.

Lecture addition: Economic Interpretation of VFI

- **Initial guess** $v^0 = 0$: One-period model (no future utility). Agent consumes all output today.
- **First iteration** Tv^0 : Two-period model. Optimize today/tomorrow, world ends after tomorrow.
- **Convergence**: Each iteration extends horizon by 1 period. $n \rightarrow \infty \rightarrow$ infinite-horizon value function.

Recording: 00:56:48, 00:58:31

1.7. Using the value function analytically

Assume u , f , and v differentiable; optimal choices interior. Substitute $c = f(k) + (1 - \delta)k - k'$ into Bellman:

$$v(k) = \max_{k'} [u(f(k) + (1 - \delta)k - k') + \beta v(k')], \quad (1.19)$$

over feasible k' .

- **FOC for k'** :

$$-u'(c) + \beta v'(k') = 0 \Rightarrow u'(c) = \beta v'(k'). \quad (1.20)$$

- **Envelope condition**: differentiate w.r.t. k , holding optimal k' fixed:

$$v'(k) = u'(c)[f'(k) + 1 - \delta]. \quad (1.21)$$

- **Combine**: apply envelope condition next period $\rightarrow$ usual Euler equation:

$$u'(c_t) = \beta v'(k_{t+1}) = \beta u'(c_{t+1})[f'(k_{t+1}) + 1 - \delta]. \quad (1.22)$$

Same condition as sequence problem. At corners, use KKT conditions.

1.8. Using the value function numerically

Need numerical representation of $v(k)$.

- Simplest approach: restrict capital to finite grid

$$K_N = \{k^1, k^2, \dots, k^N\} \subset K. \quad (1.23)$$

- Approximate value function by vector

$$\mathbf{v} = (v_1, v_2, \dots, v_N)^\top \in \mathbb{R}^N, v_i \approx v(k^i). \quad (1.24)$$

One entry per capital grid point.

Algorithm 1.8.1 (Value function iteration (VFI)):

1. **Choose grid** $K_N = \{k^1, \dots, k^N\}$. For each pair (i, j) , compute

$$c_{ij} = f(k^i) + (1 - \delta)k^i - k^j. \quad (1.25)$$

Let J_i contain choices j with $c_{ij} \geq 0$ and finite $u(c_{ij})$. Each J_i must be nonempty.

2. **Initial guess:** any $\mathbf{v}^0 \in \mathbb{R}^N$, e.g. all zeros. Set iteration $m = 0$ and tolerance $\varepsilon > 0$.
3. **Bellman update:** for every $i = 1, \dots, N$,

$$v_i^{m+1} = \max_{j \in J_i} [u(c_{ij}) + \beta v_j^m]. \quad (1.26)$$

Use the old vector for all updates: $\mathbf{v}^{m+1} = T_N \mathbf{v}^m$.

4. **Convergence check:**

$$\Delta_m = \max_{1 \leq i \leq N} |v_i^{m+1} - v_i^m|. \quad (1.27)$$

If $\Delta_m \leq \varepsilon$, stop and return $\hat{\mathbf{v}} = \mathbf{v}^{m+1}$. Otherwise set $m \leftarrow m + 1$ and repeat step 3.

5. **Recover approximate policy:** for each i , choose

$$j_i \in \arg \max_{j \in J_i} [u(c_{ij}) + \beta \hat{v}_j]. \quad (1.28)$$

Set $k'(k^i) = k^{j_i}$ and $c(k^i) = c_{ij_i}$.

Why it converges: T_N is a β -contraction in the sup norm. **Theorem 1.6.1** $\rightarrow$ unique grid fixed point $\mathbf{v}_N^*$; **Algorithm 1.8.1** converges from any initial vector. At stopping, value-error bound:

$$\|\hat{\mathbf{v}} - \mathbf{v}_N^*\|_\infty \leq \frac{\beta}{1 - \beta} \Delta_m \leq \frac{\beta}{1 - \beta} \varepsilon. \quad (1.29)$$

This controls iteration error; grid approximation error remains.

Lecture addition: Convergence in General Equilibrium

- CMT guarantees convergence for individual decision problems (value functions).
- Heterogeneous GE: convergence of **distribution** of states (assets, income, human capital) required.
- Distribution = higher-dimensional object; may have multiple fixed points → more complex than single-agent VFI.

Recording: 01:14:16, 01:15:18

Lecture addition: Curse of Dimensionality

- Grid-based VFI suffers from **curse of dimensionality**: n state variables → M^n grid points (exponential growth).
- Large n (e.g., 50) → computationally intractable.
- **Frontier approaches**: Sparse grids; mean field games [ASR uncertain: Ben Moll]; heuristic trust in convergence when rigorous proofs unavailable.

Recording: 01:15:51, 01:16:53

Lecture addition: Complexity vs. Reality

- Model complexity driven by **expectation side**: agents anticipate others' actions → complex fixed-point problems.
- Evidence: real people not “fully forward-looking” as standard models assume.
- **Responses**: (1) Develop computational tools for complex models; (2) Simplify models to match cognitive complexity (behavioral/limited rationality).

Recording: 01:17:25, 01:19:00

Bibliography

Ljungqvist, L., & Sargent, T. J. (2018). *Recursive Macroeconomic Theory* (4th ed.). MIT Press. <https://mitpress.mit.edu/9780262038669/recursive-macroeconomic-theory/>^o